Joint Inversion of Magnetotelluric and P-Wave Travel Time Data at Mount St. Helens Volcano Using a Variation of Information Constraint



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Motivation

Mt. St. Helens, USA



- MT and seismic tomography are both methods sensitive to melt zones in volcanic areas ➔ Joint inversion
  - Testing the Variation of Information coupling in joint inversions with these two methods
    - Mount St. Helens is a very well-studied volcano:
      ➔ Comparison of results with published models
      ➔ Many existing data sets

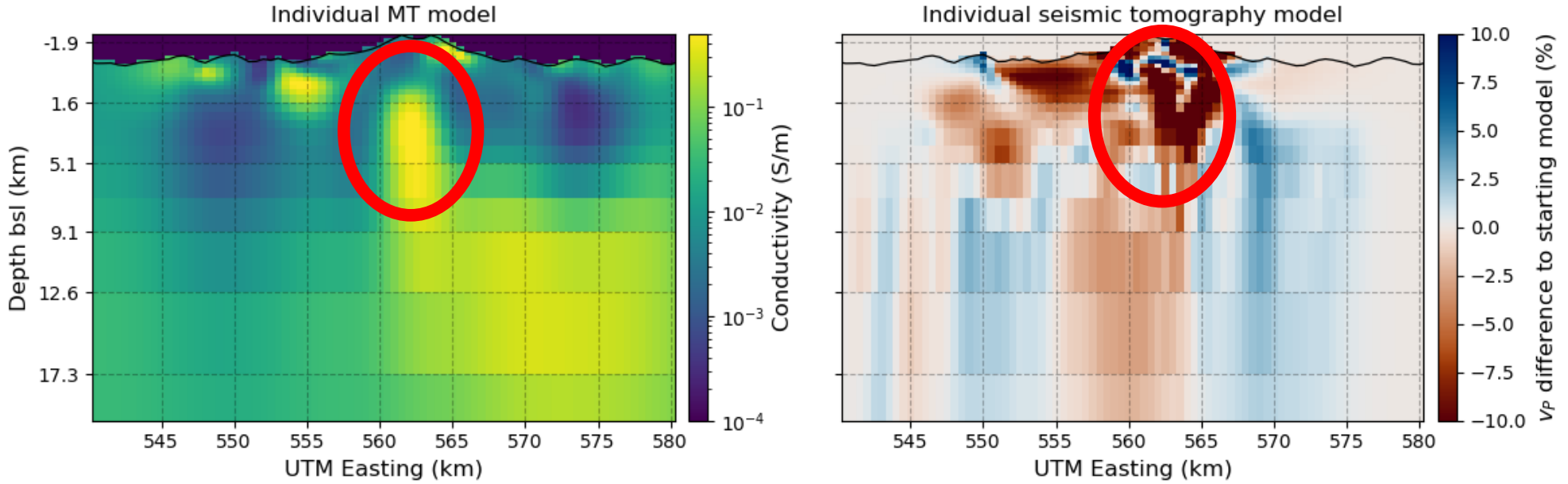
Data used here:

- MT data from Hill et al. (2009)
- P-wave travel time data from Waite & Moran (2009) and Ulberg et al. (2020)

Individual Inversions of MT and seismic travel time data

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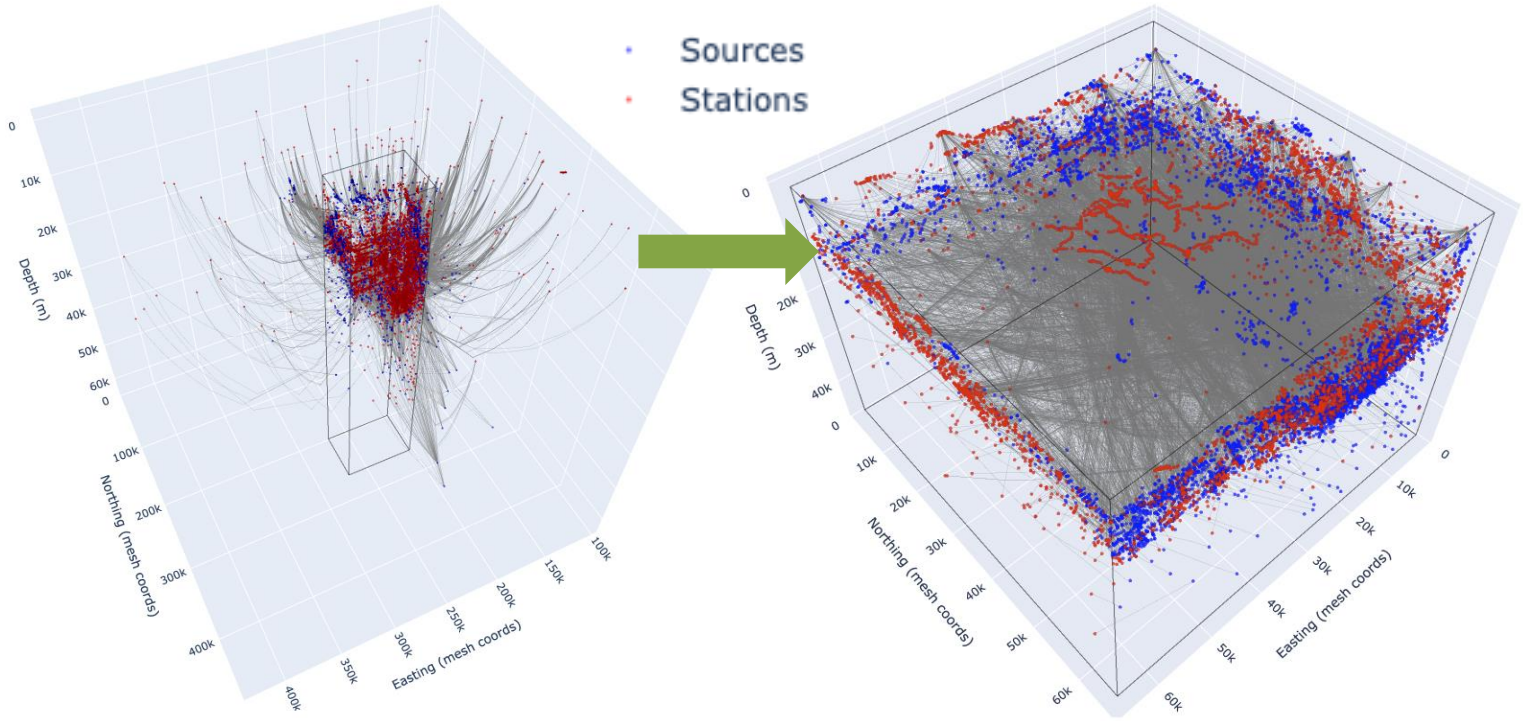
Inversions with the inversion framework *jif3d* [Moorkamp et al., 2011]



Condensing the seismic data to the MT mesh extent

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Seismic station and source spatial distribution has much larger scale than MT mesh extent ➔ needs to be more compatible



- Projection of seismic sources and receivers from outside the area of interest (AOI) onto the AOI boundary
- Forward modeling and subtraction of travel times from rays outside the AOI

➔ same mesh extent as MT,
but still dense ray coverage inside AOI

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Joint Inversion with a Variation of Information Constraint

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Variation of Information (VI) is a measure from Information Theory

- Based on the Shannon Entropy:
  H describes the uncertainty associated with a random variable.

$$\mathrm{H(X)} = -\sum_{i=1}^{n} p(x_i)\ \log p(x_i)$$

- Closely related to Mutual Information:
  I describes the information that two distributions share.

$$\mathrm{MI(X;Y)} = \sum_{x\epsilon X}\sum_{y\epsilon Y} p(x,y)\log\frac{P(x,y)}{p_X(x)p_Y(y)}$$

$$\mathrm{VI(X,Y)} = \mathrm{H(X)} + \mathrm{H(Y)} - 2\,\mathrm{MI(X;Y)}$$

➔ VI is minimized, when there is much information shared between two distributions

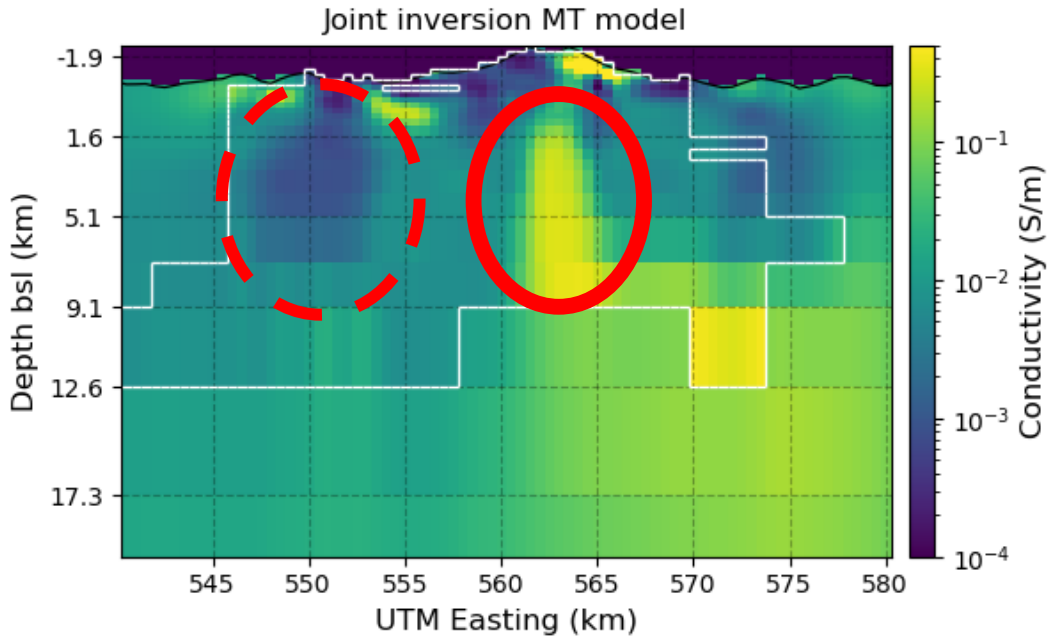
Objective function: $\Phi(\sigma, s) = \Phi_{d,MT} + \Phi_{d,Tomo} + \Phi_{Reg,\sigma} + \Phi_{Reg,s} + \Phi_{VI}$

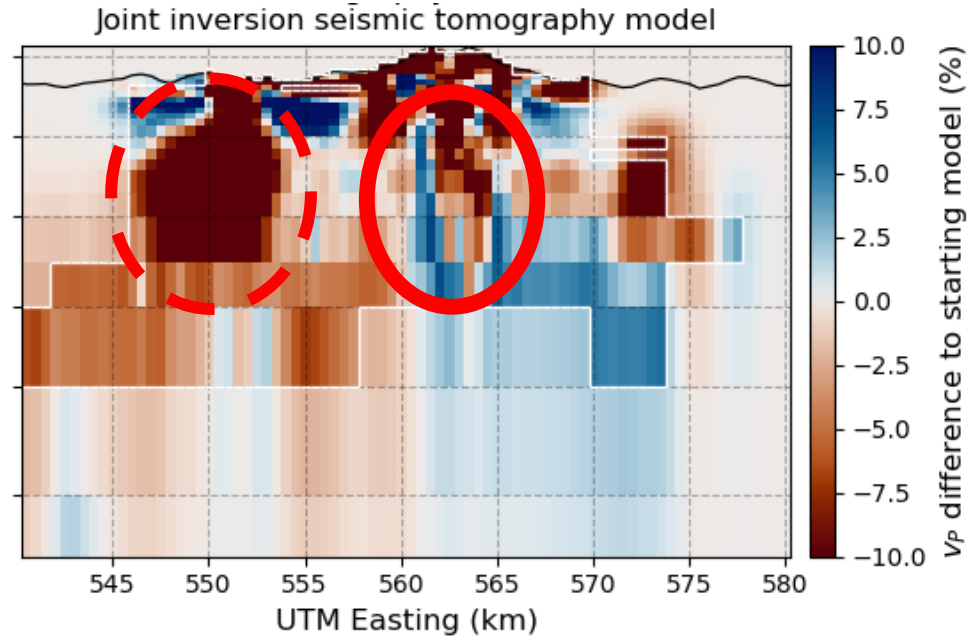
[Moorkamp, 2021]

Investigating the plumbing system of Mount St. Helens

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Joint inversion with Variation of Information (VI) as coupling constraint:





Investigating the plumbing system of Mount St. Helens

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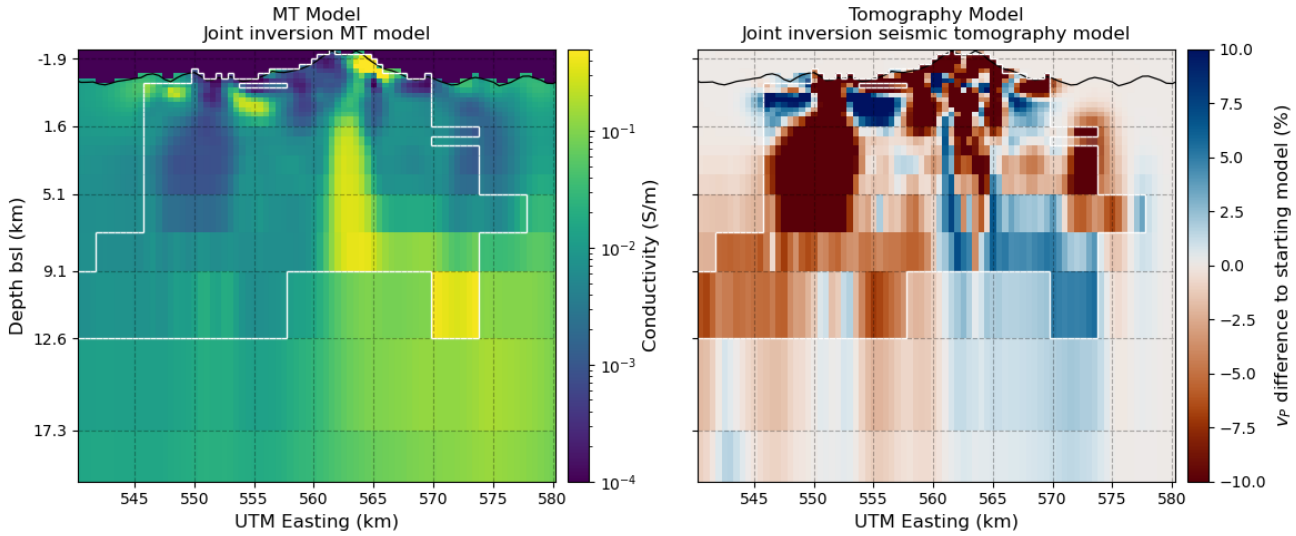
Joint inversion with Variation of Information (VI) as coupling constraint:

- VI functions like a regularization
  → it seeks a model similar to that of the other physical property
- VI enforces a one-to-one correspondence between physical properties

Interesting: Where is the constraint violated?

➔ This model region must be constrained by the data

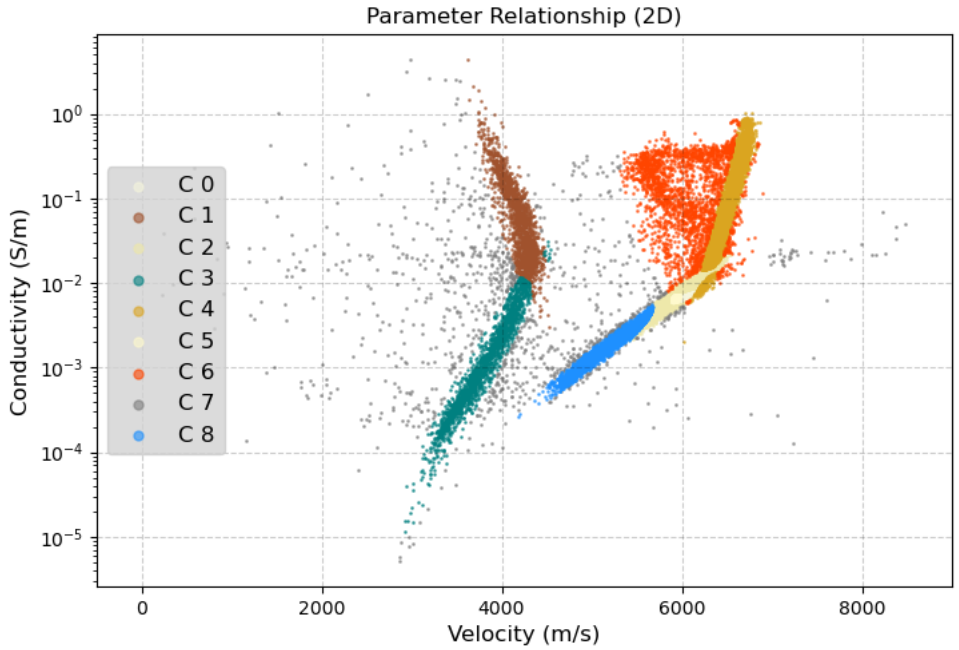
➔ All data have sufficient sensitivity in this region, otherwise the violation of the constraint would not occur

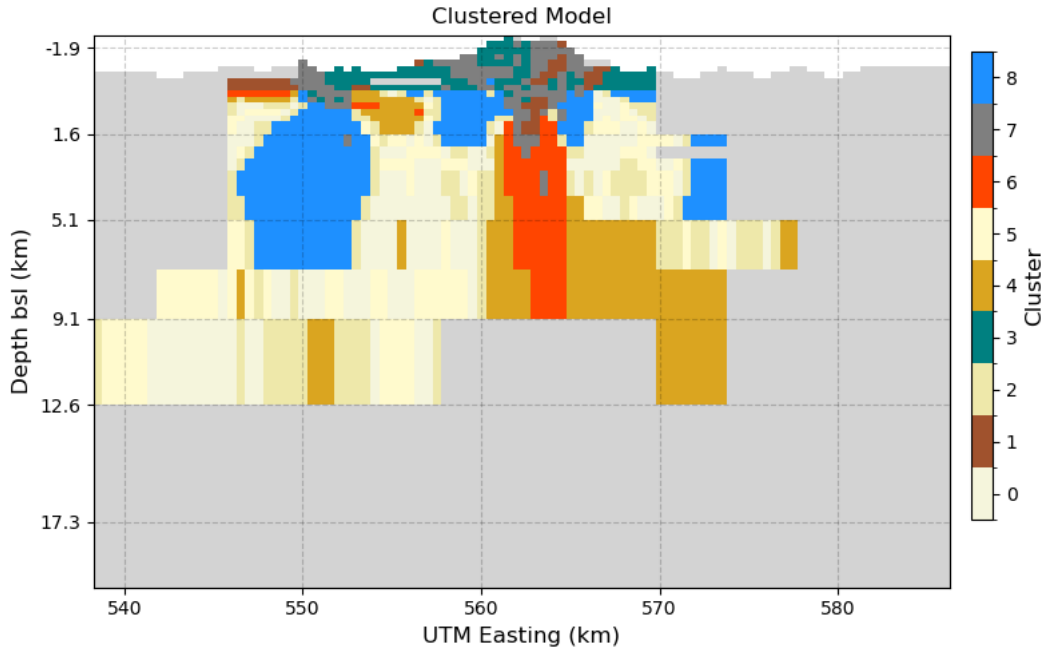


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Parameter relationship:
Clustering with Gaussian Mixture Models





Investigating the plumbing system of Mount St. Helens

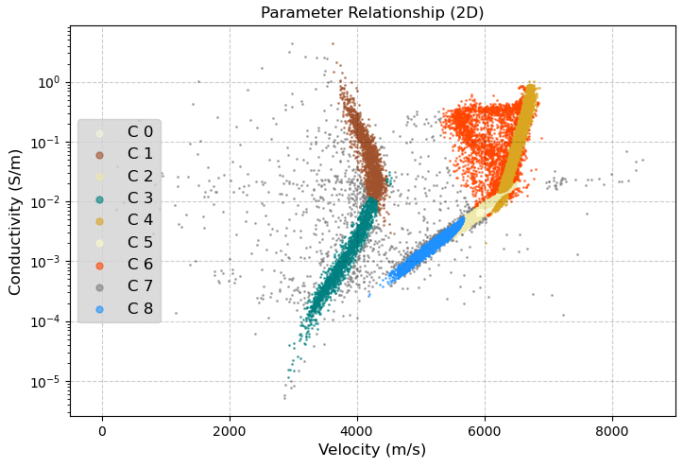
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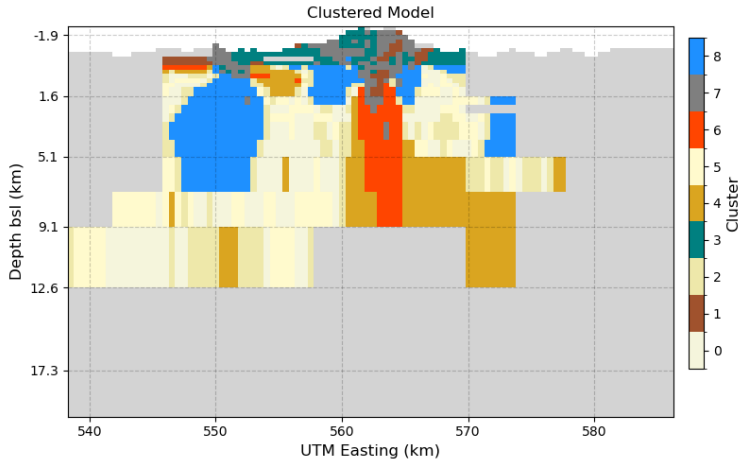
Parameter relationship: Clustering with Gaussian Mixture Models

VI is a method for investigation correlations and their violations:

- Dark orange region: Violation of the one-to-one constraint of VI
- It corresponds to a region previously identified as zone of partial melt

➔ Further investigations could include estimation of melt fraction from seismic velocity and conductivity





References

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